Mobility-Aware Cooperative Caching in IoVs Based on Secure Asynchronous Federated and Deep Reinforcement Learning

Xuefang Nie, *Member, IEEE*, Chen Wang, Tianqing Zhou, *Member, IEEE*, Qiangqiang Zhou, *Member, IEEE*, Xusheng Zhu, *Graduate Student Member, IEEE*, and Jiliang Zhang, *Senior Member, IEEE*

***Abstract*—Edge content caching of Internet of Vehicles (IoVs) is a key technology for alleviating backhaul strain and reducing access latency. To protect the privacy of vehicular users, Federated learning (FL) is employed by sharing vehicles' local models instead of data. However, vehicles may leave the coverage range of serving node before completing the local model training. To enhance model aggregation efficiency, asynchronous Federated learning (AFL) is employed, which allows asynchronous aggregation without waiting for all vehicles to update their local models. In practice, the local models are susceptible to malicious tampering during the global aggregation process. To solve this problem, we propose a secure AFL (SAFL) framework by incorporating a *Z*-score-based weight detection method within AFL. Moreover, To improve caching efficiency and adapt to the highly dynamic IoV environments, we introduce an innovative proactive caching approach by combining a conditional variational autoencoder and generative adversarial network to predict popular contents, thereby improving the cache hit ratio. Additionally, based on the prediction results of popular content, we optimize intelligent decision-making using multiagent deep reinforcement learning (DRL) to reduce the content transmission delay. Extensive simulations are performed based on real-world datasets and experimental results demonstrate that the proposed SAFL and multiagent DRL hybrid technique outperforms other baseline approaches.**

***Index Terms*—Asynchronous Federated learning (AFL), content caching, deep reinforcement learning, generative adversarial networks, popular content prediction.**

Received 24 January 2025; accepted 17 February 2025. Date of publication 21 February 2025; date of current version 9 June 2025. This work was supported in part by the Natural Science Foundation of Jiangxi Province of China under Grant 20232BAB202019 and Grant 20232ACB212005; in part by the National Natural Science Foundation of China under Grant 61961020, Grant 624B2094, and Grant 62262030; in part by the National Key Research and Development Program of China under Grant 2021YFB3300900; and in part by Graduate Innovation Special Foundation of Jiangxi Province of China under Grant YC2024-S408. *(Corresponding authors: Xuefang Nie; Chen Wang.)*

Xuefang Nie, Chen Wang, and Tianqing Zhou are with the School of Information and Software Engineering, East China JiaoTong University, Nanchang 330013, China (e-mail: xuefangnie@163.com; wangchenecjtu@163.com; zhoutian930@163.com).

Qiangqiang Zhou is with the School of Software, Jiangxi Normal University, Nanchang 330027, China (e-mail: qiang@jxnu.edu.cn).

Xusheng Zhu is with the Department of Electronic Engineering, Shanghai JiaoTong University, Shanghai 200240, China (e-mail: xushengzhu@sjtu.edu.cn).

Jiliang Zhang is with the College of Information Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: zhangjiliang1@mail.neu.edu.cn).

Digital Object Identifier 10.1109/JIOT.2025.3544368

I. Introduction

A. Background

With the rapid proliferation of intelligent transportation systems (ITS), connected vehicles and autonomous vehicles have become critical components to improve safety and efficiency in ITS [1]. These vehicles generate astonishing amounts of data due to the demand for real-time, computation-intensive applications, such as traffic management and mobile entertainment. This surge has spotlighted the limitations of traditional cloud computing frameworks, where data processing occurs in distant data centers, resulting in unacceptable latency and network bandwidth tension [2], [3]. Vehicular edge computing (VEC) becomes a strategic solution that allocates computing resources to the edge nodes (ENs), including base stations (BSs), roadside units (RSUs), and vehicles, in Internet of vehicles (IoVs). Meanwhile, mobile edge computing (MEC) and content caching has become an important paradigm, where popular contents cached at the ENs to effectively relieve backhaul strain and content access latency [4], [5], [6].

Although mobile edge caching can significantly improve quality of service for delay-sensitive content delivery, highly dynamic topology and unstable communications raise new challenges for caching decisions in IoVs [7]. Moreover, due to the limited caching capacities of ENs, it is essential to design an effective caching strategy and save the corresponding contents that are interesting to vehicular users (VUs) [8]. The traditional method utilizes reactive caching strategies to cache content that depends on previously requested content. However, the vehicle mobility results in highly dynamic changes in content popularity [9]. As a result, previously cached content may expire quickly, making reactive caching strategies unsuitable for VEC scenarios. Therefore, it is critical to cache popular content at the appropriate EN in advance based on the predicted results. That is, mobility-aware cooperative caching (MACC) should be conducted before the arrival of VU requests. Fortunately, MACC can be achieved through machine learning (ML) methods, which can effectively extract dynamic features from VU data, thereby better predicting popular content [10], [11], [12], [13]. In practice, the VU data generally contains personal information, and VUs are unwilling to share directly their private information with

2327-4662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

others. Consequently, collecting and training VU raw data at ENs is not realistic.

Federated learning (FL) is a promising ML technology, which can protect users' personal information by sharing vehicles' local models instead of direct raw data [14], [15], [16]. In conventional FL, the shared global model is updated periodically through globally aggregating all local models, while each vehicle performs training according to VU data and uploads to the connected RSU [17], [18]. However, owing to vehicle's high mobility, vehicle may leave the coverage range of the connected RSU before completing the local model updates, which will significantly degrade the accuracy of the predicted content popularity [19]. Fortunately, asynchronous FL (AFL) can perform asynchronous aggregation. That is, the global model can be aggregated immediately based on AFL without waiting for all local models to complete updates.

B. Motivations and Contributions

Recently, researchers have made some progress in mobility-aware cooperative edge caching, but implementing effective edge content caching in highly dynamic IoVs still faces three key issues:

1) *Model Security:* The ENs in dynamic IoVs are vulnerable to malicious attacks, which may result in model parameters being maliciously tampered during the asynchronous aggregation phase. Therefore, it is necessary to enhance the model security. However, there is no work considering the model security based on the AFL framework.
2) *Prediction Model:* The popular contents cached in ENs are easily outdated quickly due to the mobility of vehicles. To improve caching efficiency and adapt to the highly dynamic IoV environments, proactive caching should be achieved. Therefore, an accurate model to predict popular content for proactive caching should be explored. The most recent study that the authors utilized context-aware adversarial autoencoder (C-AAE) model to predict popular content [11]. However, this model is not suitable for intricate IoVs and is likely to encounter the model collapse problem in large scale MACC scenarios. Therefore, a more accurate model for predicting popular content for dynamic IoVs is highly necessary.
3) *Edge Intelligence:* To improve cache hit ratio (CHR) and reduce content transmission delay (CTD), the popular contents should be placed in local RSU, neighboring RSU and vehicle according to prediction results [20]. However, the diversity of caching contents may result in the size of popular content exceeding the caching capacities of ENs. The complex content requests and high mobility of vehicles raise new issues for caching decisions in time-varying IoVs. To fully utilize edge caching resources and maximize the advantages of content prediction, it is essential for determining effectively where the popular contents are caching and replacing in ENs by edge artificial intelligence (AI), especially for cooperative caching scenarios of vehicle and RSU. Deep reinforcement learning (DRL) has solid cognitive abilities and can tackle dynamic control problems and is suitable for sequential decision making in time-varying and high-dimensional environments [21]. It is necessary to determine caching decisions by exploiting DRL.

Motivated by the challenges outlined, this article proposes a novel MACC strategy to enhance the CHR and reduce CTD in highly dynamic IoVs. Specifically, we first present a secure AFL (SAFL) framework that incorporates Z-score-based weight detection (WDZ) within AFL to ensure model security. To further enhance cooperative caching efficiency and adapt to dynamic environments, we introduce an innovative proactive caching approach that combines a conditional variational autoencoder and generative adversarial network (CVAE-GAN) to predict popular content, thereby improving CHR. Additionally, to fully leverage caching resources and maximize the benefits of content prediction, we optimize the caching strategy for cooperative scenarios of vehicle and RSU by integrating SAFL with multiagent DRL. The main contributions of this article are summarized as follows.

1) We propose a SAFL framework to safeguard the privacy of VUs and enhance model security in dynamic IoV environments. Specifically, we introduce a WDZ-based security check within conventional AFL for asynchronous aggregation of the global model to prevent the local models from malicious tampering. Simulation results demonstrate that, under complex malicious attack scenarios, our proposed SAFL framework significantly improves the security of the global model and enhances CHR performance.
2) To improve caching efficiency and adapt to the highly dynamic IoV environments, we introduce an innovative proactive caching approach that integrates the CVAE-GAN model within SAFL to capture the underlying patterns and dynamic characteristics between VUs and popular content, thereby enhancing the accuracy of content popularity prediction. Unlike conventional models, the proposed model generates older samples through iterative training on historical tasks. By combining these historical samples with new data, we can retrain the classifier to ensure that all samples can be recognized by the final model. As a result, the CVAE-GAN model effectively addresses the issue of model collapse, enabling robust predictions even when data is limited. Simulation results demonstrate that the proposed model significantly improves CHR compared to the recent AE and C-AAE models.
3) To fully utilize edge caching resources and maximize the benefits of content prediction within SAFL, we further optimize the edge caching strategy for cooperation between vehicle and RSU in highly dynamic IoVs. The cooperative caching decision process is first formulated as a Markov decision process (MDP). We then develop a multiagent DRL-based approach, called asynchronous advantage actor–critic (A3C), within SAFL framework to address this MDP problem, optimizing deep neural network (DNN) to obtain the optimal policy through asynchronous gradient descent. Simulation

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

results demonstrate that the proposed multiagent A3C approach outperforms the baseline schemes, achieving significant performance gain in terms of CHR and CTD in highly dynamic IoVs.

C. Organization

The remainder of this article is structured as follows. The related works on edge content caching for IoVs are presented in Section II. The system architecture of edge content caching is described in Section III. The SAFL framework and popular content prediction based on CVAE-GAN model are presented in Section IV. Section V discusses cooperative caching decisions by multiagent DRL-based method to reduce CTD. Additionally, Section VI presents the simulation results and Section VII concludes this article.

II. Related Work

We summarize the relevant works for mobility-aware edge content caching in this section. These works are reviewed in three categories, including cooperative ENs type for content caching, AFL for predicting popular content and DRL for edge caching decision.

A. Cooperative EN Type for Content Caching

Edge caching is an important technology that can significantly reduce the latency of content transmission, since VUs can fetch directly contents from ENs in IoVs. Cooperative edge caching among ENs have attracted widespread research attention in recent years. Many current studies on content caching strategies can be classified according to where the contents are cached. The cooperative EN type for content caching includes vehicular caching [22], UAV caching [23], [24], and RSU caching [25], [26], [27].

For vehicular caching, Yao et al. [22] investigated the content caching policy based on mobility prediction to cache in vehicles in hot-spot areas. The authors employed a partial matching approach to predict the probability of vehicles entering different hot-spot areas based on their trajectories. For UAV caching, Liu et al. [23] and Qin et al. [24] proposed a DRL-based scheme aimed at jointly optimizing the content caching and task computation to reduce the computation delay and enhance the CHR of UAV-assisted vehicular networks (VNs), respectively. For RSU caching, Wang et al. [25] designed a content update algorithm based on Markov clustering to optimize the caching strategy of RSUs in each cluster, thereby improving the CHR of requested contents in vehicular ad hoc networks (VANETs). The cooperative caching among RSUs was further studied in [26], where the authors utilized the surrounding function-characteristic of vehicles to evaluate service preferences, while considering the transportation correlation of vehicle trajectories. The proactive cooperative caching was studied in [27]. The authors designed a popular content prediction algorithm based on spatial-temporal correlation and further investigated a multiagent reinforcement learning (RL) policy for content cached at RSUs in VNs. In summary, there is little research on the cooperative caching among BSs, RSUs, and vehicles in highly dynamic IoVs.

B. AFL for Popular Content Prediction

Designing caching replacement strategies is primarily influenced by content popularity [28]. Some conventional rule-based edge cache strategies, for example, most frequently used [29] and least recently used [30], can only approximately predict content popularity and cannot adapt to large-scale and high dynamic VUs. Recently, many researches on edge caching replacement strategies within FL framework have been presented [10], [11], [12], [13], [14]. Zhao and Nakhai [10] investigated a caching replacement strategy that integrates FL with deep Q learning for edge networks, aiming to reduce system cost and increase CHR. To enhance CHR of edge caching, Yu et al. [11] incorporated C-AAE model within FL to predict the content popularity. Due to the low training efficiency based on FL, Wu et al. [12] proposed a AFL approach to address the model training problem, and incorporated auto-encoder (AE) to improve the accuracy of predicting popular content. Similarly, Jiang et al. [13] developed a stacked AE model within AFL framework to predict the content popularity in the Internet of Things (IoTs). Recently, Qian et al. [14] exploited proactive video caching based on AFL for MEC networks. The authors employed an AE model to extract hidden features and designed a caching algorithm based on a filtering model to identify popular videos. To date, model security solutions based on AFL for edge caching scenarios remain unexplored.

C. DRL for Edge Caching Decision

With the resurgence of AI technologies, DRL-based methods have been widely applied for edge content caching decisions [21], [31], [32], [33], [34], [35]. Compared with conventional optimization algorithms, DRL-based methods perform excellently in sequential decision making especially in dynamic decisions under uncertainty.

Ye et al. [31] proposed an intelligent framework for computing and caching resource allocation tailored to the dynamic features of IoVs. Moreover, the authors introduced an A3C-based DRL approach to jointly optimize offloading and caching, demonstrating the superiority of DRL approach. Zhang et al. [32] considered content caching scenarios and applied dueling DDQN (D3QN)-based DRL to optimize multimodel content caching. Wu et al. [12] developed a Dueling DQN (DuDQN) method to perform caching decisions and reduce CTD. The MACC scheme, which combines AFL and DRL, is further investigated in [13], where the authors designed a double DQN (DDQN) algorithm to make edge caching decisions. Wu et al. [33] extended the caching strategy to the time-varying device to device (D2D) offloading scenario, and presented a recommendation-enabled caching strategy integrates supervised learning with DRL. Wang et al. [34] considered multinode cooperative caching scenarios, applied Newton's cooling law to calculate content popularity, and developed a DQN-based DRL algorithm to maximize caching efficiency of ENs, including

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

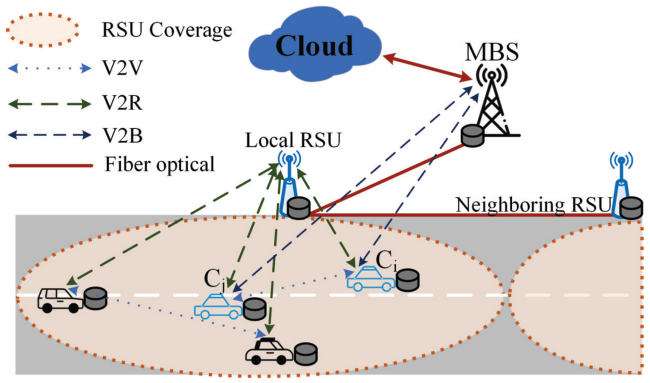


Fig. 1. Illustration of network architecture.

CHR and energy efficiency, in the IoTs. In practice, the mobility of terminal devices significantly influences caching efficiency. Therefore, Bi and Zhao [21] employed long short-term memory to predict content popularity based on historical information and developed a DQN algorithm to optimize access paths for content downloading from BS and cluster vehicles in VNs. Building upon this, Zhang et al. [35] explored joint content caching and computation offloading for MEC systems, and developed a two-tier DQN algorithm to address service caching and cooperative computing.

Through our review of cooperative content caching, we have identified a key limitation of the current AFL framework in preventing malicious tampering, highlighting the need to enhance model security. Moreover, prediction accuracy is crucial for proactive caching, yet existing popularity prediction models are not well-suited for highly dynamic IoV environments. Additionally, based on predicted content popularity, intelligent decisions for cooperative caching between vehicle and RSU within SAFL are crucial for improving caching efficiency. However, effective caching decision-making that integrates multiagent DRL with SAFL in highly dynamic IoVs remain unexplored. This article addresses these challenges and fills these gaps.

III. System Model

This section introduces the system architecture for edge content caching, including network infrastructure, communication model, and mobility model of vehicles. The notation definition is listed in Table I.

A. System Architecture

A three-tier IoV is considered for urban environments that contains a macro BS (MBS), a local RSU, a neighboring RSU, and some vehicles. Fig. 1 shows the system architecture of IoVs. The MBS in the first tier is deployed at the center of IoVs. RSUs in the middle tier are deployed on side of the road. Each RSU is located at the coverage range of MBS. Vehicles in the bottom tier are driving in the coverage range of RSUs. In this article, we assume that each vehicle corresponds to a single vehicle user (VU). However, our model can also be applied to scenarios where multiple users are present within a single vehicle. In such cases, we assume that all users within the vehicle share a unified content preference. Each mobile vehicle carries a substantial amount of local data, which directly reflects the personal information of the VUs. This data includes access request information, ratings for all available contents, and context information of VUs, such as gender, age, and identity (ID) numbers, etc. A high rating of a content means that it is greater interest to users. When the rating of the content is 0, it indicates that the content has not been requested or is not popular. The local data is utilized as the training set according to a certain percentage, while the remaining data is utilized as a testing set.

The MBS is associated with the Internet through an ideal fiber optical link. Both RSU and MBS have MEC servers with content caching and computing capabilities. In particular, each RSU has decision-making capability to determine where to answer content requests and which local cached content should be replaced. We assume that MBS has sufficient cache capacity to store all available contents, while RSUs and vehicles have limited cache capacity and can only cache small amounts of contents. The network system operates in a multitime slot manner (e.g., round). The total number of vehicles within the

TABLE I
Notation Definition

Notation	Definition
$\mathbb{C}^r$	Set of vehicles in local RSU
N^r	The number of vehicles in local RSU
X	VU's rating matrix
y	Potential vector
μ_e	The mean of potential vector
σ_e^2	The variance of potential vector
Z	VU's personal information matrix
X_{recon}	VU's reconstructed rating matrix
X_{sample}	Randomly generated sample data X
y_{sample}	Randomly generated sample data y
$D(X)$	Discriminator outputs for real data
$D(X_{sample})$	Discriminator outputs for real generated data
λ_v	Mean rate of arrival
V_i^r	Velocity of the vehicle C_i^r
μ_v	Mean of velocity
σ_v^2	Variance of velocity
S	Diameter of RSU.
T_{train}	Average training time
T_{inf}	Inference time
ω^r	Global model
n_i^r	Training data.
$\mu_{\Delta\omega_{key}}$	The mean of the weight differences
$\sigma_{\Delta\omega_{key}}$	The variance of the weight differences
d_i^r	Data size of the C_i^r
$\hat{X}_i^r$	Reconstructed based on X_i^r
r	Number of FL communication rounds
T_s	Rounds of A3C algorithm
θ	Parameters of actor network
θ_u	Parameters of critic network
ψ	Parameter for entropy regularization strength

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

coverage range of the local RSU in a round r is denoted as N^r. Let the set of vehicles N^r be $\mathbb{C}^r = \{C_1^r, C_2^r, \ldots, C_i^r, \ldots, C_{N^r}^r\}$, where C_i^r is the ith $(1 \le i \le N^r)$ vehicle traveling within the coverage range of the local RSU. For each round, each vehicle first gathers the information of content requests and then fetches the corresponding contents from the neighboring vehicles, local RSU, neighboring RSU, or MBS according to different situations. Specifically,

1) Neighboring Vehicle: The local vehicle C_i^r delivers directly the contents to its VUs when the required content is cached in the local vehicle. Otherwise, the local vehicle C_i^r forwards the request to adjacent vehicle C_j^r with approximate cosine similarity formulated in (30), and the neighboring vehicle C_j^r delivers corresponding content to the local vehicle C_i^r when the requested contents are cached in the neighboring vehicle C_j^r. The vehicle C_i^r obtains the requested contents from neighboring vehicle C_j^r in this case.

2) Local RSU: The vehicle C_i^r sends the request to the local RSU if the corresponding content is not cached in vehicle C_j^r. Then vehicle C_i^r forwards the request to the local RSU and the local RSU delivers the content to the vehicle C_i^r if it caches the required content. The vehicle C_i^r obtains the corresponding contents from the local RSU in this case.

3) Neighboring RSU: The request is forwarded to the neighboring RSU if the local RSU does not cache the requested content of vehicle C_i^r. Then the neighboring RSU sends the requested content to the local RSU if it cached the corresponding content and the local RSU subsequently transfers the corresponding content to the vehicle C_i^r. Therefore, the vehicle C_i^r fetches the requested content from neighboring RSU in this case.

4) MBS: The vehicle C_i^r directly transmits a request to MBS when it is unable to obtain the requested content from neighboring vehicle, local RSU and neighboring RSU. MBS sends the requested content back to the vehicle C_i^r because it has all available contents. Therefore, the vehicle C_i^r will directly fetch the requested content from the MBS in this case.

B. Communication Model

The communications between vehicle and MBS (V2B) or RSU (V2R) employ the orthogonal frequency division multiple access technique, while the multichannel allocation is adopted for the communications from vehicle to vehicle (V2V) using the dedicated short range communication (DSRC) technique. The orthogonal spectrum allocation is adopted between MBS and RSU. Thereby the interference is not take into account in this model. A more realistic communication model is beyond the scope of this article and is left for future work. The content transmission from the local RSU to neighboring RSU is carried on through an ideal fiber optical link. During a round, each vehicle maintains same communication model, and the communication model may change for different rounds. According to the Shannon theorem, the communication rate $R_{i,j}$ from vehicle C_i to vehicle C_j in round r is calculated as [36]

$$R_{i,j}^r = B_v \log_2\left(1 + \frac{p_v h_{ij}^r\left(\mathrm{dis}\left(C_i^r, C_j^r\right)\right)}{\sigma_c^2}\right) \tag{1}$$

where B_v is the available DSRC bandwidth from 5.850 to 5.925 GHz, p_v is the transmit power of vehicle C_i, σ_c^2 is the noise power, $h_{ij}^r(\mathrm{dis}(C_i^r, C_j^r))$ is the channel gain of the V2V link, where $\mathrm{dis}(C_i^r, C_j^r)$ is the distance from the vehicle C_i to vehicle C_j in round r. The channel gain of V2V link in round r is modeled as

$$h_{ij}^r\left(\mathrm{dis}\left(C_i^r, C_j^r\right)\right) = \alpha_{ij}\left(\mathrm{dis}\left(C_i^r, C_j^r\right)\right) g_{ij}^r \tag{2}$$

where $\alpha_{ij}(\mathrm{dis}(C_i^r, C_j^r))$ is the large-scale fading consisting of pathloss and shadowing. The path-loss can be computed by $128.1 + 37.6\log_{10}(\mathrm{dis}(C_i^r, C_j^r)$ [12] and the shadowing follows a log-normal distribution, g_{ij}^r is the small-scale Rayleigh fading that follows an exponential distribution with unit mean.

The rate $R_{s,i}$ between the local RSU and vehicle C_i in round r can be calculated as

$$R_{s,i}^r = B_s \log_2\left(1 + \frac{p_s h_{si}^r\left(\mathrm{dis}(S, C_i^r)\right)}{\sigma_c^2}\right) \tag{3}$$

where B_s is channel bandwidth, p_s represents the transmission power of RSU, $h_{si}^r(\mathrm{dis}(S, C_i^r)$ is the channel gain, and S represents the local RSU. Similarly, the transmission rate $R_{b,i}$ between MBS and vehicle C_i in round r can be computed by

$$R_{b,i}^r = B_m \log_2\left(1 + \frac{p_m h_{mi}^r\left(\mathrm{dis}(M, C_i^r)\right)}{\sigma_c^2}\right) \tag{4}$$

where B_m is channel bandwidth, p_m is the transmit power of MBS, $h_{mi}^r(\mathrm{dis}(M, C_i^r))$ is the channel gain between MBS and vehicle C_i in round r, and M represents MBS.

C. Mobility Model of Vehicles

Assuming that all vehicles are traveling in the same direction, and the vehicle's arrival in the coverage range of the local RSU follows Poisson distribution with arrival rate λ_v. There are no vehicles entered or left the coverage range of the local RSU during a round. Therefore, the total number of vehicles located in the coverage range of RSU remains unchanged within a round. The mobility features of vehicle, including velocity and position, are fixed within a round, and may change at start each round. The velocities of all vehicles follow an independent identically distributed truncated Gaussian distribution [37]. Let V_i^r be the velocity of vehicle C_i^r. The velocities of vehicles driving in the coverage range of RSU in a round r are denoted by $\{V_1^r, V_2^r, \ldots, V_i^r, \ldots, V_{N^r}^r\}$. Let the probability density function (PDF) of V_i^r denote $f(V_i^r)$, which is calculated as

$$f(V_i^r) = \begin{cases} \dfrac{e^{-\frac{1}{2\sigma^2}(V_i^r - \mu_v)^2}}{\sqrt{2\pi\sigma_v^2}\left(erf\left(\frac{V_x - \mu_v}{\sigma_v\sqrt{2}}\right) - erf\left(\frac{V_m - \mu_v}{\sigma_v\sqrt{2}}\right)\right)}, & V_i \le V_i^r \le V_x \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

where V_x and V_m represent the maximum and minimum threshold of velocity V_i^r, respectively. $erf(V_i^r - \mu_v/\sigma_v\sqrt{2})$ is the Gauss error function of V_i^r with mean μ_v and variance σ_v^2. Let D_i^r denote the traversed distance of C_i from current position to the entrance of local RSU in round r. According to [38], the traversed distance D_i^r can be computed by

$$D_i^{r+1} = D_i^r + V_i^r T(r) \tag{6}$$

where $T(r)$ represents the duration time of V_i^r in round r.

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

IV. Secure Asynchronous Federated and Deep Reinforcement Learning

In this section, we present a mobility-aware content caching scheme based on SAFL and DRL. We first introduce a SAFL framework to protect VU data and train model, where a WDZ method is proposed to enhance security of the local model. Then we introduce a popular content prediction method based on CVAE-GAN model to improve the predicted accuracy. Furthermore, we design a DRL-based algorithm to determine cooperative edge caching decision-making according to the results of popular content prediction.

A. Secure AFL-Based Model Training

AFL-based model training process is illustrated in Fig. 2. In synchronous FL, the global aggregation performed in the local RSU must wait for all vehicles to complete the upload of their local models. Subsequently, each vehicle will download the updated model from the local RSU. Considering the high-mobility of vehicles, we employ the AFL approach to train the local model, where each vehicle trains the local model iteratively by local data for each round, and uploads updates to the local RSU. For each round, the local RSU can immediately aggregate the global model without waiting for all vehicles to complete the upload of the local models. We present SAFL considering both the mobility of vehicle and model security in this section. The steps of SAFL are described as follows.

1) Vehicle Selection: Some vehicles may leave the coverage range of the local RSU quickly due to the high mobility of vehicle and the small coverage range of RSU. To ensure vehicle has the opportunity to participate in AFL training, Those vehicles with sufficient remaining time within the coverage range of RSU are first selected. Therefore, each vehicle first sends its position and velocity to the associated RSU, and then RSU selects vehicle based on the remaining time that the vehicle is still within the coverage range of the local RSU. The remaining time $T^t_{r,i}$ of vehicle C^r_i within the coverage range of the local RSU is computed by

$$T^t_{r,i} = (L_s - D^r_i)/V^r_i \tag{7}$$

where L_s represents the coverage range of RSU, and V^r_i denotes the velocity of vehicle C^r_i. The remaining time $T^t_{r,i}$ should exceed the sum of the long-term average training time T_{train} and inference time T_{inf}. That is to say, if $T^t_{r,i} > T_{\text{train}} + T_{\text{inf}}$, the vehicle C^r_i is selected.

2) Model Download: After completing the asynchronous aggregation of the global model, RSU initializes the global model based on CAVE-GAN. Subsequently, vehicles download the global model and then iteratively train and update the local models.

3) Local Training: Vehicles first obtain the global model ω^r from the local RSU and set it as their initial local model. Subsequently, each vehicle updates local model after completing iteratively training. Let $n^r_{i,k}$ be the local training data. For each iteration k, the local empirical loss function for vehicle C^r_i is defined as

$$f(\omega^r_{i,k}) = \frac{1}{|n^r_{i,k}|} \sum_{X \in n^r_{i,k}} \mathcal{L}(\omega^r_{i,k}; \mathbf{X}) \tag{8}$$

where $|n^r_{i,k}|$ represents the number of training data $n^r_{i,k}$, $\mathcal{L}(\omega^r_{i,k}; \mathbf{X})$ is the loss function of data point $\mathbf{X}$ condition on the local model $\omega^r_{i,k}$, which is further described in formula (29). In order to reduce the deviation between the local model $\omega^r_{i,k}$ and global model ω^r and thus ensuring the algorithm convergence, the regularized function $g(\omega^r_{i,k})$ can be computed by

$$g(\omega^r_{i,k}) = f(\omega^r_{i,k}) + \frac{\lambda}{2} \left\| \omega^r - \omega^r_{i,k} \right\|^2 \tag{9}$$

where λ is the regularization parameter.

Some vehicles are unable to successfully upload the local model due to the training time delay. We called these vehicles as stragglers and degrade the accuracy of the global model and algorithm convergence [11], [12], [13]. For the sake of clarity, let $\nabla g(\omega^r_{i,k})$ denote the gradient of regularized function $g(\omega^r_{i,k})$, while the local gradient of stragglers are called as the delayed local gradient. To tackle with the issue, in current round r, we first aggregate all local gradient including the delayed local gradient. Therefore, the local gradient is expressed as

$$\nabla \zeta^r_{i,k} = \nabla g(\omega^r_{i,k}) + \alpha \nabla g^d_i \tag{10}$$

where α is the decay coefficient and ∇g^d_i is the delayed local gradient. Notably, $\nabla g^d_i = 0$ when C^r_i successfully uploads in previous round. In the next iteration, the local model can be expressed as

$$\omega^r_{i,k+1} = \omega^r - \eta^r_d \nabla \zeta^r_{i,k} \tag{11}$$

where $\eta^r_d = \eta_d \max\{1, \log(r)\}$ represents learning rate in round r, and η_d is the initial learning rate.

Subsequently, the local model finishes the kth iteration and vehicle C^r_i samples training data again to turn into the next iteration and continues to loop until the maximum iteration threshold is reached. Then the vehicle C^r_i completes the local training and uploads model ω^r_i to the local RSU. The loss function of the global model can be expressed as

$$\mathcal{L}(\omega^r) = \sum_{i=1}^{N^r} \frac{d^r_i}{D^r} f(\omega^r_i) \tag{12}$$

where d^r_i represents the local data size of vehicle C^r_i and D^r denotes the overall size of the local date in the selected vehicles. Our target is to construct the model ω^* to minimize $\mathcal{L}(\omega^r)$, i.e.,

$$\omega^* = \arg\min_{\omega^r} \mathcal{L}(\omega^r). \tag{13}$$

4) Upload Model: As illustrated in Fig. 2, vehicles completes the training of local model, the updated local model is uploaded to local RSU for the asynchronous aggregation of the global model.

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

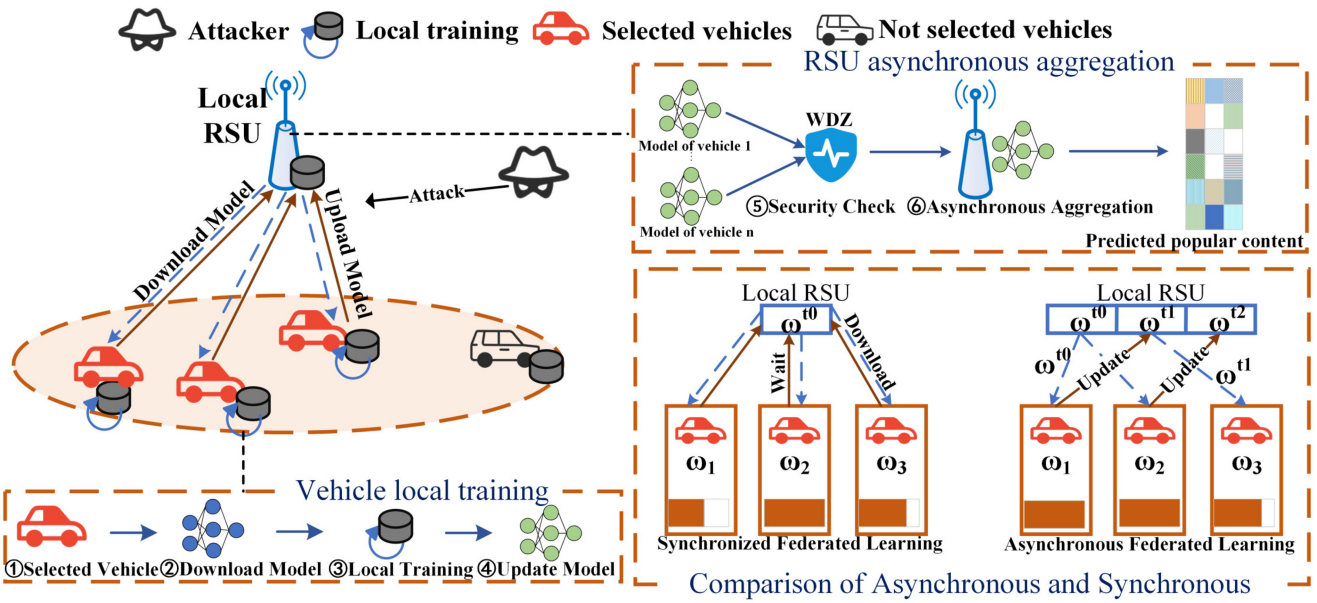


Fig. 2. SAFL based mobility-aware model training process.

5) WDZ-Based Security Check: Wireless channel transmission for uploading local models is vulnerable to malicious attacks and eavesdropping. To address this issue, we propose a WDZ scheme using the Z-score method to perform security check at the local RSU, and thus improving the security of local model. The WDZ-based method is a numerical measurement technique used in statistics, which is calculated based on the group's mean and standard deviation. A high or low Z-score indicates a significant deviation between the data point and the mean. Therefore, the data point will significantly deviate the mean if the local model is subjected to network attacks. In the SAFL framework, The local model has been uploaded, the local RSU first calculate the weight difference between the local model ω_i^r in current round and ω_i^{r-1} in the previous round. The weight difference $\Delta\omega_i^d$ can be calculated as

$$\Delta\omega_i^d = \omega_i^r - \omega_i^{r-1}. \tag{14}$$

Based on the weight difference, the mean $\mu_{\Delta\omega_d}$ and standard deviation $\sigma_{\Delta\omega_d}$ of the weight difference are calculated as, respectively

$$\mu_{\Delta\omega_d} = \frac{1}{K}\sum_{i=1}^{K}\Delta\omega_i^d \tag{15}$$

and

$$\sigma_{\Delta\omega_d} = \sqrt{\frac{1}{K-1}\sum_{i=1}^{K}(\Delta\omega_i^d - \mu_{\Delta\omega_d})^2} \tag{16}$$

where K represents the number of weights of the local model. Based on the calculated mean and standard deviation, it is considered abnormal when the following condition is satisfied

$$\left\|\mu_{\Delta\omega^d}\right\| > \rho^r \times \sigma_{\Delta\omega^d} \tag{17}$$

where ρ^r is a threshold for anomaly detection in round r. Once an abnormality is identified during the security check, the RSU will discard the local model ω_i^r of vehicle C_i^r for asynchronous aggregation in current round. In practice, various types of attacks may be encountered during the weight uploading process. To enhance the adaptability of the WDZ mechanism, this article proposes a dynamic threshold adjustment mechanism. This mechanism adjusts the threshold, denoted as ρ, based on the anomaly rate ξ in each round, thereby better accommodating variations in node behavior differences and balancing false positives and false negatives. During each round of asynchronous aggregation, the anomaly rate ξ is first calculated. It represents the ratio of the number of local models identified as anomalous to the total number of local models in the current round, i.e.,

$$\xi^r = \frac{N_{\text{anom}}^r}{N_{\text{total}}} \tag{18}$$

where N_{anom}^r and N_{total} represent the number of local models deemed anomalous and the total number of local models, respectively. Based on the anomaly rate ξ in each round of aggregation, the threshold ρ is dynamically adjusted to maintain a balance between false positives and false negatives. The specific adjustment formula can be described as

$$\rho^{r+1} = \begin{cases} \rho^r + 0.1, & \text{if } \xi^r > 0.3 \\ \rho^r - 0.1, & \text{if } \xi^r < 0.1 \\ \rho^r, & \text{otherwise} \end{cases}. \tag{19}$$

When the anomaly rate exceeds 30%, it indicates that too many models are being flagged as anomalous, which may lead to false positives. In this case, the threshold ρ should be increased to reduce the misclassification of normal nodes. Conversely, when the anomaly rate is below 10%, it suggests that the detection may not be sensitive enough to identify anomalous nodes. In such cases, the threshold ρ should be decreased to enhance detection sensitivity. Through this dynamic adjustment mechanism, the threshold ρ is adaptively adjusted during each training round, thereby reducing the

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

likelihood of false positives. This ensures the stability and performance of the global model, even in the face of complex and dynamically changing node behavior.

6) Asynchronous Aggregation: The global model is updated immediately in RSU by asynchronous aggregating local models of vehicles. Unlike the conventional synchronous FL, the AFL can perform asynchronous aggregation of global model without waiting for all local models to complete uploading. In the AFL, the local RSU can immediately update the global model as soon as the local model update arrives. According to the asynchronous aggregation [12], when the local model of vehicle C_i is successfully uploaded, the global model can be updated by

$$\omega^{r+1} = \omega^r - \eta_d^r \frac{d_i^r}{D^r} \nabla \zeta_i^r. \tag{20}$$

In the AFL, we take the vehicle mobility into account and formulate the updated global model as

$$\omega^{r+1} = \omega^r - \eta_d^r \psi_i \frac{d_i^r}{D^r} \nabla \zeta_i^r \tag{21}$$

where ψ_i represents the weight of the asynchronous aggregation for C_i^r. We consider the traversed distance of vehicle C_i^r within the coverage of the local RSU and the transmission delay from local RSU to vehicle C_i^r. If the vehicles C_i^r travels a longer distance within the RSU coverage range, it stays connected for a longer duration, allowing it to accumulate more training time and perform additional iterations. As a result, its model should be given greater weight during asynchronous aggregation to enhance the overall model accuracy. Based on the wireless transmitting rate and the remaining distance within the coverage range of RSU, the aggregation weight of vehicle C_i^r can be calculated as [12]

$$\psi_i = \tau \frac{(L_s - D_i^r)}{L_s} + (1-\tau) \frac{R_{s,i}^r}{\max\limits_{i \in N^r}(R_{s,i}^r)} \tag{22}$$

where τ is the balancing weight factor between the position weight and the transmission rate weight.

Up to now, the model training is completed. The algorithm of SAFL is illustrated in detail in Algorithm 1.

7) Convergence Analysis: We conducted a convergence analysis on the proposed SAFL algorithm considering the vehicle selection and asynchronous aggregation. Several definitions and assumptions were first provided as follows.

Definition 1 (L-Smoothness): The local loss function f is L-smooth with $L > 0$, if for $\forall n_1, n_2$

$$f(n_1) - f(n_2) \le \langle \nabla f(n_2), n_1 - n_2 \rangle + \frac{L}{2} \|n_1 - n_2\|^2. \tag{23}$$

Definition 2 (μ-Strongly Convexity): The loss function f is μ-strongly convex with $\mu > 0$ if for $\forall n_1, n_2$,

$$f(n_1) - f(n_2) \le \langle \nabla f(n_2), n_1 - n_2 \rangle + \frac{\mu}{2} \|n_1 - n_2\|^2. \tag{24}$$

Definition 3 (Q-Locally Dissimilarity): The aggregated gradient $\nabla \zeta_i^r$ exhibits Q-locally dissimilarity with $Q > 0$ if

$$\mathbb{E}\left\|(\nabla \zeta_i^r)\right\|^2 \le Q^2 \|\nabla \mathcal{L}(\omega)\|. \tag{25}$$

Algorithm 1 Mobility-Aware and Popular Content Prediction Based on SAFL With CVAE-GAN

The local RSU execution:
Input: The position, velocity and the transmission rate of vehicles, and the balancing weight factor β.
Initialize the global model ω^r;
  for each round r **do**
    $T_{r,i}^t = (L_s - D_i^r)/V_i^r$
    If $T_{r,i}^t > T_{train} + T_{inf}$ **then**
      Select vehicles to participate in the AFL for model training
  for each selected vehicle **do**
    Download the current global model
    $\omega_i^r \leftarrow$ Vehicle Updates(ω^r, i)
    Upload local updated model ω_i^r to the local RSU
    The local RSU receive the updated model ω_i^r and perform anomaly detection on the weights using the Z-score method based on (17)
    Calculate the weight ψ_i based on (22)
    Calculate the updated model ω^{r+1} based on (21)
Return the global model ω^{r+1}
Vehicle execution:
Input: Download ω^r, initialize learning rate η_d, decay coefficient α, and regularize parameter λ
Each vehicle calculates $\eta_d^r = \eta_d \max\{1, \log(r)\}$ based on the local initialization learning rate η_d
Vehicle Updates(ω^r, i)**:**
Input: the global model ω^r
for each epoch **do**
  Train local model based on CVAE-GAN model and calculate loss function based on Eq. (8)
  Compute the regularized function based on Eq. (9)
  Update iteratively the model ω_i^r according to Eq. (11)
Return the updated model ω_i^r

Assumption 1: The global loss function $\mathcal{L}(\omega)$ is bounded and there exists $\varepsilon > 0$ if for $\forall \omega$, $\mathbb{E}(\nabla \zeta_i^r) \le \|\nabla \mathcal{L}(\omega)\|$ and $\nabla \mathcal{L}(\omega)^\top \mathbb{E}(\nabla \zeta_i^r) \ge \varepsilon \|\nabla \mathcal{L}(\omega)\|^2$.

Based on Assumption 1, we further illustrate the convergence of the proposed SAFL using the subsequent lemmas and theorems.

Lemma 1: If $\mathcal{L}(\omega)$ is strong convex, we have

$$2\mu\Big(\mathcal{L}(\omega^r) - \mathcal{L}(\omega^*)\Big) \le \left\|\nabla \mathcal{L}(\omega^r)\right\|^2. \tag{26}$$

Theorem 1: If $\mathcal{L}(\omega)$ is L-smooth and μ-strongly convex, $\nabla \zeta_i^r$ is Q-locally dissimilar, and Assumption 1 is held. Then, let $\eta_d^r \ge (2\varepsilon d_i^r / LQ^2\, D^r) \ge \eta_d$. The SAFL will converge to the optimal solution ω^* after $R^{\max}$ rounds, i.e.,

$$\begin{aligned} &\mathbb{E}\Big[\mathcal{L}\Big(\omega^{R^{\max}}\Big) - \mathcal{L}\big(\omega^*\big)\Big] \\ &\le \big(1 - 2\mu\eta_d \rho'\big)^{R^{\max}} \Big[\mathcal{L}\Big(\omega^0\Big) - \mathcal{L}\big(\omega^*\big)\Big] \end{aligned} \tag{27}$$

where $\rho' = \varepsilon - (\eta_d^r L Q^2/2)$.

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

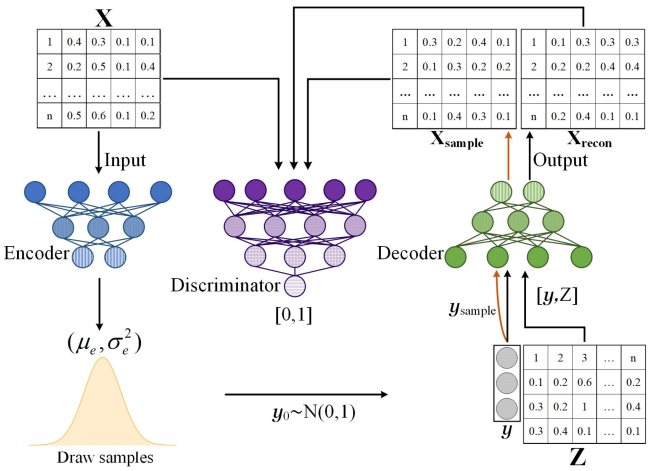


Fig. 3. CVAE-GAN model architecture.

Theorem 2: If $\mathcal{L}(\omega)$ is L-smooth and nonconvex, $\nabla\zeta_i^r$ is Q-locally dissimilar and Assumption 1 is satisfied, after $R^{\max}$ rounds, we can obtain

$$\sum_{r=0}^{R^{\max}-1} \frac{\eta_d^r}{2}\mathbb{E}\left(\left\|\nabla\mathcal{L}\left(\omega^0\right)\right\|^2\right) \leq \mathcal{L}\left(\omega^0\right) - \mathcal{L}\left(\omega^*\right) \tag{28}$$

where $\eta_d^r \leq (2\varepsilon - 1/LQ^2)$.

In summary, Theorem 1 guarantees the convergence of the SAFL algorithm, especially for the convex objective function and offers an error approximate bound for the aggregation model. Theorem 2 shows that the convergence speed of the SAFL is controllable. The detailed proofs for Lemma 1, Theorems 1 and 2 are presented in [19] and bib12. Interested researchers can further read these sources.

B. CVAE-GAN-Based Popular Content Prediction

In this section, we first introduce the architecture of CVAE-GAN model shown in Fig. 3. The CVAE-GAN-based popular content prediction is then presented.

The CVAE-GAN integrates the advantages of conditional variational autoencoders (CVAE) and generative adversarial networks (GAN). The CVAE is responsible for simulating data distribution and generating high-quality data, while the GAN is used to enhance the authenticity and diversity of the data through the adversarial process between discriminator and generator. Specifically, the CVAE employs a carefully designed encoder network to transform the input content request matrix $\mathbf{X}$ into latent variables y, with the aiming of capturing the core features of the input data. These latent variables are then fed into the decoder network, which attempts to reconstruct the original input $\mathbf{X}_{\text{recon}}$ from these compressed representations. This process aids the model in learning how to restore and generate data, ensuring that the produced content is not only diverse but also closely resembles real data. The discriminator of GAN evaluates the similarity with the actual data distribution. This adversarial training mechanism forces the decoder to continually optimize its generation process to produce more accurate and indistinguishable fake data. This not only boosts the model's generative capability but also ensures more accurate and reliable predictions. Moreover, the personal information matrix $\mathbf{Z}$ of VUs is integrated into the model, incorporating but not limited to personal information, geographical location, historical request data and temporal information. The fusion of these data enables the model to better understand and predict the types of content that specific user groups may be interested in. Therefore, the CVAE-GAN model not only predicts the content popularity but also provides highly personalized prediction in IoVs.

The training process of CVAE-GAN contains reconstruction phase and adversarial phase. In the reconstruction phase, the rating matrix $\mathbf{X}$ of VU is taken as the input vector of the encoder to obtain the potential vector y that is calculated by multiplying the sampling vector y_0 by the standard deviation σ_e^2. Furthermore, the potential vector y is fused with the personal information matrix $\mathbf{Z}$ of VU to form input matrix $[y, \mathbf{Z}]$ of the decoder network. Subsequently, the reconstructed rating matrix $\mathbf{X}_{\text{recon}}$ can be obtained. In the adversarial phase, $\mathbf{X}$, $\mathbf{X}_{\text{sample}}$ and $\mathbf{X}_{\text{recon}}$ represent real data, randomly sampled data and reconstructed data, respectively. The decoder of the CVAE also serves as the generator of GAN to produce synthetic data that is close to the real data. The discriminator can distinguish between "real" and "false" samples. Meanwhile, the discriminator can diminish the risk of mode collapse and ensure the reconstructed data is more real. During the training process, the discriminator and generator engage in a competitive interaction to continuously improve each other's performance. The loss function of the CVAE-GAN model is calculated by

$$\begin{aligned}\mathcal{L}(\omega_{i,k}^r; \mathbf{X}) = \ & \tau_1\left(-\frac{1}{2}\sum_{j=1}^{J}\left(1+\log(\sigma_{e,j}^2) - \mu_{e,j}^2 - \sigma_{e,j}^2\right)\right) \\ & + \tau_2\left(\frac{1}{2}\left(\|\mathbf{X} - \mathbf{X}_{\text{recon}}\|_2^2\right)\right) \\ & + \tau_3\left(\mathbb{E}\left[\log\left(D(\mathbf{X})\right)\right] + \mathbb{E}\left[\log\left(1 - D(\mathbf{X}_{\text{sample}})\right)\right]\right)\end{aligned} \tag{29}$$

where the first to third lines of the above formula represent the loss functions of the encoder, decoder, and discriminator, respectively. τ_1, τ_2, and τ_3 ($\tau_1 + \tau_2 + \tau_3 = 1$) are weighting parameters, J is the dimensional size of μ_e and σ_e^2, $D(\mathbf{X})$ and $D(\mathbf{X}_{\text{recon}})$ are the outputs from the discriminator network for real data and generated data, respectively.

Next, we describe the content prediction process. As shown in Fig. 4, the prediction algorithm for popular content based on CVAE-GAN model contains the following four steps.

1) Data Preprocessing: It is very difficult to distinguish whether a content is of interest to VU when its rating is 0. In the first step, we need to reconfirm the rating of each content, as described below.

Each vehicle first extracts a rating matrix data from the testing set, where the first and the second dimensions of the matrix are VU's ID and content rating, respectively. Let the rating matrix for C_r^i be $\mathbf{X}$. Then the rating matrix $\mathbf{X}$ is reconfirmed based on the CVAE-GAN model, where the matrix $\mathbf{X}$ and rating matrix $\mathbf{X}_{\text{recon}}$ are the input and output of the CVAE-GAN model, respectively. Because the rating matrix

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

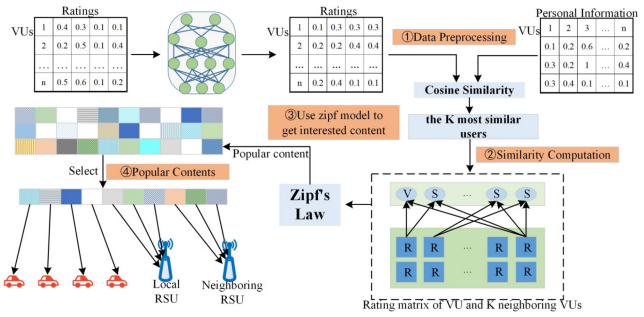


Fig. 4. Popular content prediction process.

$\mathbf{X}_{\text{recon}}$ is reconfirmed which can capture the latent features of data, and thus it can be regarded as an approximation of matrix $\mathbf{X}$. Similarly, the personal information matrix (PIM) data is also extracted from the testing set. In particular, the first dimension and second dimension of the matrix are the ID and personal information of VUs, respectively.

2) Similarity Computation: Vehicle C_i^r computes the number of nonzero ratings for each VU in matrix $\mathbf{X}_i^r$ and identifies the VUs with top m numbers as active VUs. Subsequently, the rating matrix and the PIM are combined by each vehicle to compute the cosine similarity between active VUs and other VUs. According to the calculation method of cosine similarity [12], the cosine similarity between an active VUs p and q can be calculated by

$$\begin{aligned}\sin_{p,q}^{r,i} &= \cos\left(\boldsymbol{M}_i^r(p,:), \boldsymbol{M}_i^r(q,:)\right) \\ &= \frac{\boldsymbol{M}_i^r(p,:) \cdot \boldsymbol{M}_i^r(q,:)^T}{\|\boldsymbol{M}_i^r(p,:)\|_2 \times \|\boldsymbol{M}_i^r(q,:)\|_2}\end{aligned} \tag{30}$$

where $\boldsymbol{M}_i^r(p,:)$ and $\boldsymbol{M}_i^r(q,:)$ are the vectors of the active VU p and q, respectively. $\|\boldsymbol{M}_i^r(p,:)\|_2$ and $\|\boldsymbol{M}_i^r(q,:)\|_2$ are the 2-norm of $\boldsymbol{M}_i^r(p,:)$ and $\boldsymbol{M}_i^r(q,:)$, respectively. For active VU p, vehicle C_i^r picks out K VUs with the highest similarities as its neighboring VUs. At the same time, the ratings of the K neighboring VUs can also directly reflect the preferences of VU p.

3) Interested Contents: The K neighboring VUs of each active VU are identified, the ratings of n contents are then evaluated by the K neighboring VUs. Each vehicle subsequently identifies the content with nonzero ratings as the content popularity. These contents are then sorted and processed through Zipf model, where the input for the Zipf model is the ranking of historical request quantities across the entire content repository. The Zipf model not only helps identify the most popular content overall but also helps determine the relative popularity of content among different VU groups. This method can bring more personalized content recommendations and improve user satisfaction.

4) Popular Contents: Each vehicle C_i^r selects the m most popular contents as the predicted request contents. After collecting the predicted popular contents for all vehicles, the RSU sorts these contents and selects the top F_c contents as the most popular contents. These most popular request contents are placed at vehicles, the local RSU or neighboring RSUs according to the caching decisions based on DRL, which is described as follows.

C. DRL for Edge Caching Decision

In order to fully utilize the storage capacities of RSU and vehicle, the content replacement mechanisms are necessary for edge caching with limited capacities. Each RSU is assumed to be capable of computation capability and can make the real-time cooperative edge caching decisions. Therefore, we explore intelligent edge caching decision making through DRL method to maximize the benefits of predicted popular content in IoVs. What follows, we first propose the framework of DRL and further design the intelligent algorithm based on DRL.

1) *DRL Framework:* The DRL framework for content caching is modeled as MDP, which consists of state space, action space, and reward. The DRL-based training process involves multiple slots. Specifically, RSU first observes the current state $s(t)$ and subsequently performs an action $a(t)$ with a policy π. And then, RSU obtains a reward $r(t)$ for current slot t and then observes the transited state $s(t+1)$. What follows, we further describe the state $s(t)$, action $a(t)$ and reward $r(t)$ within the DRL framework, respectively.
2) *State Space:* The current state is the instant environment, which means that the contents cooperatively cached at RSUs or vehicles. Let $s(t)$ be the current state space. The corresponding contents of $s(t)$ are sorted according to the predicted results of content popularity provided by RSU. As a result, the current state space is written as $s(t) = (s_1, s_2, \ldots, s_n)$, where s_n represents the nth most popular content.
3) *Action Space:* After obtaining a state $s(t)$ in each slot, the agent needs to determine a caching decision, i.e., action $a(t)$. During the process for edge content caching decisions, the popular contents are first sorted in descending order according to the popularity of predicted contents. Action space $a(t)$ indicates whether it is necessary to update the content cached on the RSU and vehicle. We assume that the predicted request contents that are not cached at ENs (RSUs or vehicles) form a content set $\mathcal{K}$. If the action space $a(t) = 1$, the vehicle will randomly select $m (m < n)$ popular contents from set $\mathcal{K}$ and replace the m least popular contents. Then these popular contents are sorted in a descending order relying on their popularity to turn into state $s_v(t+1)$. Similarly, RSU also randomly select m popular contents from set $\mathcal{K}$ as the cached contents within the next state $s(t+1)$. It is worth noting that the popular contents cached at both vehicles and the local RSU, as well as the neighboring RSU should be different. On the other hand, if $a(t) = 0$, the popular contents cached at vehicle will not be replaced, while RSU including the local RSU and neighboring RSU also do not need to replace their cached contents.
4) *Reward:* The reward $r(t)$ is utilized as a performance metric to evaluate the action quality with the aim of minimizing the total CTD. In practice, the CTD varies when the popular content is placed at different nodes.

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

If the requested content $f(f \in s_v(t))$ from vehicle C_i is cached in neighboring vehicle C_j, the delivery delay from neighboring vehicle C_j to vehicle C_i can be calculated as

$$d^r_{i,j,f}(t) = \frac{z_c}{R^r_{i,j}} \tag{31}$$

where z_c is the size of content data, and $R^r_{i,j}$ is the delivery rate from C_j to C_i, which is given by formula (1). If the content f of vehicle C_i is placed at the local RSU, i.e., $f \in s(t)$, the corresponding CTD can be calculated as

$$d^r_{s,i,f}(t) = \frac{z_c}{R^r_{i,j}} + \frac{z_c}{R^r_{s,i}} \tag{32}$$

where $R^r_{s,i}$ is the delivery rate between the local RSU to vehicle C_i. If the requested content is placed at neighboring RSU, i.e., $f \in s_n(t)$, the neighboring RSU first transmits the requested content to the local RSU. Subsequently, the local RSU forwards the corresponding content to vehicle C_i. Therefore, the CTD can be computed by

$$d^r_{n,i,f}(t) = \frac{z_c}{R^r_{i,j}} + \frac{z_c}{R^r_{s,i}} + \frac{z_c}{R_{ss}} \tag{33}$$

where R_{ss} is the delivery rate from the neighboring RSU to the local RSU, which is assumed to be a constant rate for the optical link. If the requested content is neither cached in the neighboring vehicle nor RSUs, the MBS will send the content f to C_i, then the delivery delay can be calculated as

$$d^r_{b,i,f}(t) = \frac{z_c}{R^r_{i,j}} + \frac{z_c}{R^r_{s,i}} + \frac{z_c}{R_{ss}} + \frac{z_c}{R^r_{b,i}} \tag{34}$$

where $R^r_{b,i}$ is the delivery rate from MBS to C_i. For the sake of clarity, we calculate delay reward that vehicle C_i fetches the corresponding content f in time slot t as

$$d^r_{i,f}(t) = \begin{cases} e^{-\lambda_1 d^r_{i,j,f}}, & f \in s_v(t) \\ e^{-\lambda_2 d^r_{s,i,f}}, & f \in s(t) \\ e^{-(\lambda_2 d^r_{s,j,f} + \lambda_3 d^r_{n,i,f})}, & f \in s_n(t) \\ e^{-\lambda_4 d^r_{b,i,f}}, & f \notin \{s_v(t), s(t), s_n(t)\} \end{cases} \tag{35}$$

where $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$ and $\lambda_1 < \lambda_2 < \lambda_3 \ll \lambda_4$. Let N^r and M^r_i, respectively represent the number of all vehicle and the number of the corresponding requested contents from vehicle C^r_i, then reward $r(t)$ can be expressed as

$$r(t) = \sum_{i=1}^{N^r} \sum_{f=1}^{M^r_i} d^r_{i,f}(t). \tag{36}$$

The overall objective of caching decision-making is to find the optimal cooperative caching policy π^*, thereby minimizing $\mathcal{P}(\pi)$. The optimal policy π^* can be written as

$$\pi^* = \arg\min \mathcal{P}(\pi). \tag{37}$$

where $\mathcal{P}(\pi)$ is the long-term average discounted reward, which can be computed by

$$\mathcal{P}(\pi) = \mathbb{E}_\pi \left[\sum_{t=1}^{T_s} \gamma_d^{t-1} r(t) \right] \tag{38}$$

where $\gamma_d \in [0, 1]$ is a discount factor.

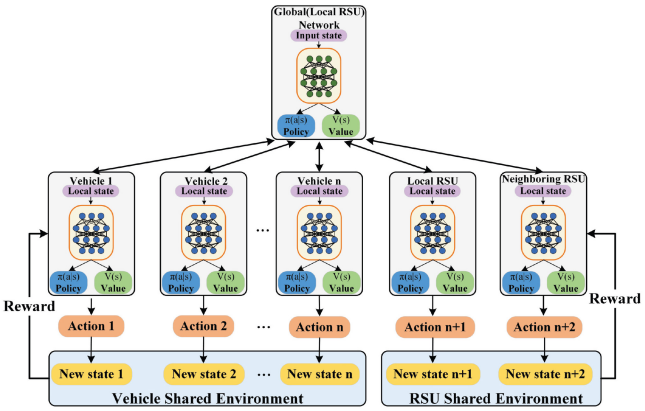


Fig. 5. Multiagent A3C-based asynchronous model training process.

2) *DRL Algorithm:* Widely used DRL algorithms, such as DuDQN [12], DDQN [13], and D3QN [32], adopt an experience replay mechanism, requiring the agent to utilize the off-policy learning approaches and update the network parameters relying on the data obtained from previous policies. Asynchronous RL can optimize DNN through the gradient descent approach and combine multiple RL algorithms. Among them, the A3C-based asynchronous RL shows the best performance in finding the optimal strategy across various continuous and discrete action spaces [31].

The multiagent A3C enhances both computer resource utilization and model training efficiency by leveraging multiple threads for synchronous training. Specifically, each thread works as an agent and multiple agents can interact with environments in parallel to learn and compute the policy gradient. Afterwards, each thread can obtain updated parameter from the global network. Unlike the other baseline algorithms, the multiagent A3C does not require randomly selecting historical samples from experience replay pool to disturb correlated data, thereby saving cache space on ENs. In this system, we propose a multiagent A3C-based algorithm where vehicles and RSUs function as asynchronous worker threads, collaboratively optimizing the model. The main RSU serves dual roles: serving as a global node for aggregating and optimizing the model, while also engaging in local training with neighboring RSUs and vehicles. It aggregates the local models from associated vehicles and neighboring RSUs, optimizes the global policy, and then distributes the updated parameters to all participating nodes. This multiagent collaborative and asynchronous training strategy enables the multiagent A3C algorithm to achieve outstanding real-time performance and system robustness in highly dynamic environments. The process of the multiagent A3C-based model training is illustrated in Fig. 5.

Given the state value $\mathcal{V}(s_t; \theta_u)$ and state space s_t in slot t, the agent selects an action based on policy $\pi(a_t|s_t; \theta)$ and turns into the next state s_{t+1}, and subsequently obtains R_t. The state value function in the multiagent A3C algorithm can be calculated as

$$\begin{aligned} \mathcal{V}(s_t; \theta_v) &= \mathbb{E}[G_t \mid s = s_t, \pi] \\ &= \mathbb{E}\left[\sum_{m=0}^{\infty} \gamma_d^m r(t+m) \mid s = s_t, \pi \right] \end{aligned} \tag{39}$$

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

where $G_t = \sum_{m=0}^{\infty} \gamma_d^m r(t+m)$ is the cumulative reward for state space s_t. The multiagent A3C algorithm employs a m-step reward to update parameters, which is given by

$$R_t = \sum_{i=0}^{m-1} \gamma_d^i r(t+i) + \gamma_d^m \mathcal{V}(s_{t+m}; \theta_u) \tag{40}$$

where $r(t+i)$ denotes the instant reward obtained in slot $t+i$. The upper limit of m is $t_{\max}$. The value function and the policy function may be updated when the final state or $t_{\max}$ are reached. In order to enhance the efficiency and training stability, an advantage function is incorporated in the multiagent A3C algorithm, which can be written as

$$\mathcal{A}(s_t, a_t; \theta, \theta_u) = R_t - \mathcal{V}(s_t; \theta_u) \tag{41}$$

where θ and θ_u are the actor network (AN) and critic network (CN) parameters. The advantage function $\mathcal{A}_t$ can enhance decision-making capabilities since it does not overestimate or underestimate the action. Based on $\mathcal{A}_t$, the loss function $\mathcal{L}_\pi(\theta)$ of actor network is calculated by

$$\mathcal{L}_\pi(\theta) = \log \pi(a_t|s_t; \theta)\big(R_t - \mathcal{V}(s_t; \theta_u)\big) + \psi \mathcal{H}\big(\pi(s_t; \theta)\big) \tag{42}$$

where the entropy term $\mathcal{H}(\pi(s_t; \theta))$ of policy π is an objective function, which is used to encourage training exploration and avoid premature convergence, and ψ is the strength of entropy regularization. The loss function $\mathcal{L}_u(\theta)$ of the CN can be written as

$$\mathcal{L}_u(\theta) = (R_t - \mathcal{V}(s_t; \theta_u))^2. \tag{43}$$

Next, the policy of the AN is updated based on the accumulation gradients, which can be calculated by

$$\begin{aligned} d\theta \leftarrow d\theta + \nabla_{\theta'} \log \pi\big(a_t \mid s_t; \theta'\big)\big(R_t - \mathcal{V}(s_t; \theta_v)\big) \\ + \psi \nabla_{\theta'} \mathcal{H}\big(\pi\big(s_t; \theta'\big)\big) \end{aligned} \tag{44}$$

Similarly, the policy of the CN is updated by

$$d\theta_u \leftarrow d\theta_u + \frac{\partial (R_t - \mathcal{V}(s_t; \theta_u))^2}{\partial \theta_u'}. \tag{45}$$

The parameters θ' and θ_u' of AN and CN are updated through minimizing loss functions of AN and CN, respectively. The relevant parameters of global neural network are subsequently updated by θ and θ_u, which is calculated by

$$\theta = \theta - \delta_1 d\theta, \quad \theta_u = \theta_u - \delta_2 d\theta_u \tag{46}$$

where δ_1 and δ_2 represent the learning rates of the AN and CN, respectively.

The multiagent A3C-based algorithm is summarized as Algorithm 2.

Algorithm 2 MultiAgent A3C Cooperative Caching Decision

Input: discount factor γ_d, the parameters of the AN θ, minimizing the loss functions of actors θ', learning rates of the AN δ_1, the parameters of the CN θ_u, minimizing the loss functions of CN θ_u', learning rates of the CN δ_2.
Output: The optimal cooperative caching strategy.
Initialization:
Initialize θ and θ_u for the global AN and CN;
Initialize global shared and thread-specific counter $T \leftarrow 0$ and $t \leftarrow 1$, respectively;
Initialize the parameters θ' and θ_u' for the thread-specific AN and CN;
Initialize γ_d, δ_1, δ_2, T_s and $t_{\max}$
Iteration:
While $T < T_s$ **do**
  for each worker **do**
    Initialize the global gradients: $d\theta = 0, d\theta_u = 0$
    Synchronize the parameters θ and θ_u of the global AN and CN with the parameters θ' and θ_u' of each thread's AN and CN
    Receive the state space s_t
    for $t < t_{max}$ **do**
      Execute a_t based on policy $\pi(a_t \mid s_t; \theta')$.
      Transition to state s_{t+1} after receiving reward r_t
      $t \leftarrow t + 1$
    end for
    $R_t = \begin{cases} 0, & \text{terminal state,} \\ V(s_t, \theta_v u'), & \text{other state.} \end{cases}$
    for $i \in \{t-1, t-2, ..., 1\}$ **do**
      $R \leftarrow R(s_t, a_t) + \gamma_d R$
      Compute the accumulation gradient θ' based on Eq. (42)
      Compute the accumulation gradient θ_u' based on Eq. (43)
    end for
    Asynchronously update parameters θ and θ_u based on Eq. (44)
    $T = T + 1$
  end for
end while

V. Simulation Results

In this section, we evaluate the performance of the proposed MACC optimization scheme, which combines SAFL and multiagent A3C (referred to as SFMA3C).

A. Simulation Settings

We consider that the VEC scenario in urban areas consists of a single MBS, multiple RSUs, and some vehicles driving on the road covered by the RSUs. Assuming that the coverage range of each RSU is assumed to be 500 m, the initial cache capacity of each vehicle is 20 contents, while each RSU possesses an initial cache capacity of 50 contents. Unless otherwise specified, the density of vehicles is assumed to be 15 vehicles/km, and the other simulation parameters settings are the same as [13]. The simulation tools contain Python 3.8 and Pytorch 2.1. The dataset utilized in this article is from MovieLens [39], which contains about 1 million ratings from 6 040 anonymous users for 3 883 movies. The ratings varies from 0 to 5. The dataset also contains users' personal information about the users, such as gender, age, and

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

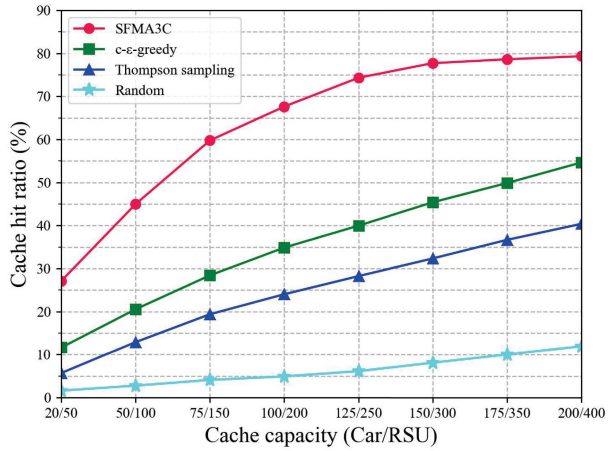


Fig. 6. CHR varies with different cache capacities.

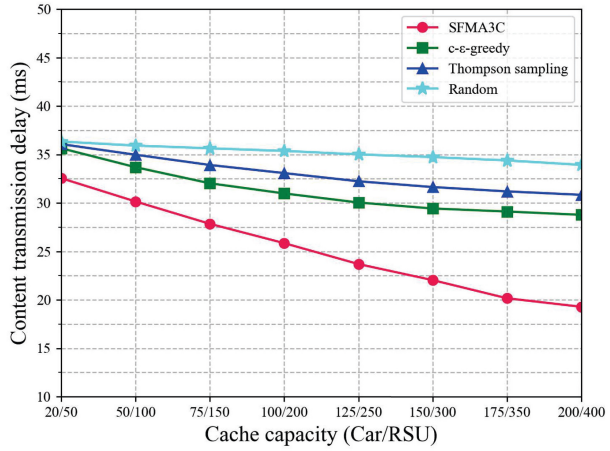


Fig. 7. CTD varies with different cache capacities.

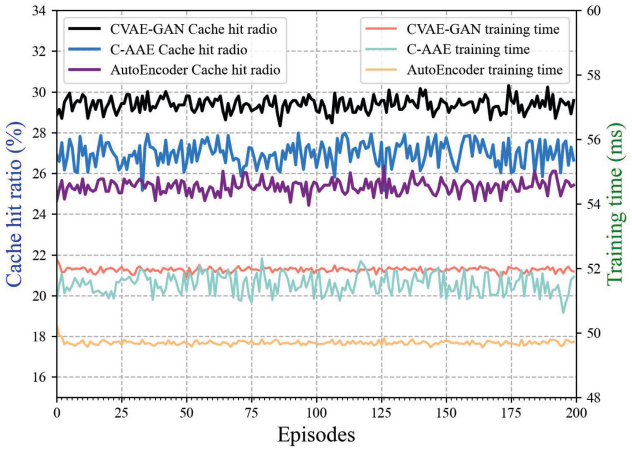


Fig. 8. CHR and training time of different predicted models.

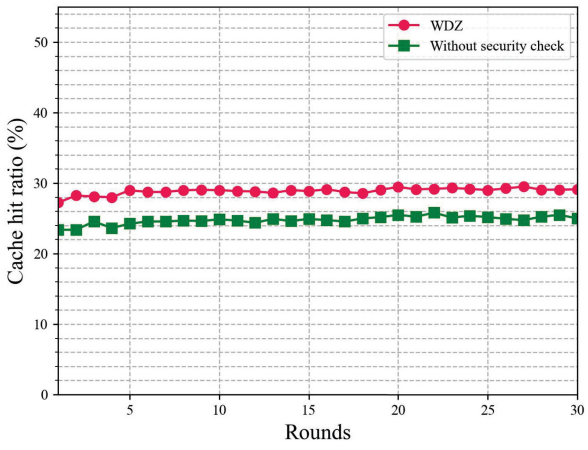


Fig. 9. Comparison of CHR between based on WDZ and without security check under random malicious attacks.

occupation and so on. We assume that the ratings for movies equates to content request for that preferred content.

B. Performance Evaluation

We evaluate the proposed SFMA3C-based MACC scheme in terms of CHR and CTD performance metrics. The CHR is used to reflect the ratio that vehicles fetch successfully requested contents from ENs [12]. If the requested content is cached in the EN, the vehicle can directly fetch the requested content from the EN, which is called a cache hit. Otherwise, it is called as a cache miss. Therefore, the CHR can be calculated as follows: CHR = cache hits/(cache hits + cache misses).

The CTD performance reflects the average delivery time for every vehicle to obtain the requested contents, which is calculated as the ratio of the overall transmission time of the requested contents to the number of vehicles.

We compare the proposed MACC scheme based on SFMA3C with the following baseline schemes described as follow.

1) *n-ϵ-greedy:* Selecting the n most frequently requested contents with a probability 1-ϵ, and selecting randomly n with a probability ϵ to cache in vehicles and RSUs. In the simulation, $\epsilon = 0.1$.
2) *Thompson Sampling:* Replacing the caching contents in vehicles and RSUs based on the probability of cache hits and cache misses in the previous round, and selecting the n contents with the highest probability to cache in vehicles and RSUs.
3) *Random:* Selecting n contents randomly to cache in the vehicles and RSUs.
4) *FedAVG:* Federated averaging (FedAVG) is a typical synchronous FL scheme that adopts a weighted average method and requires all vehicles to complete the update of the local models.
5) *AE:* The typical AE model [13] and C-AAE model [11] are generally employed to predict the interested contents of vehicle.
6) *DRL:* The typical Dueling DQN [12], DDQN [13], and D3QN [32] methods employed in current works are the baseline schemes to validate the effectiveness of the proposed SFMA3C method.

We first evaluate the CHR performance of different schemes with the variation of cache capacities of ENs (vehicle and RSU). From Fig. 6, it can be seen that the CHR of all schemes rise with the cache capacities of ENs. This is due to the fact that ENs with a larger capacity can cache more contents,

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

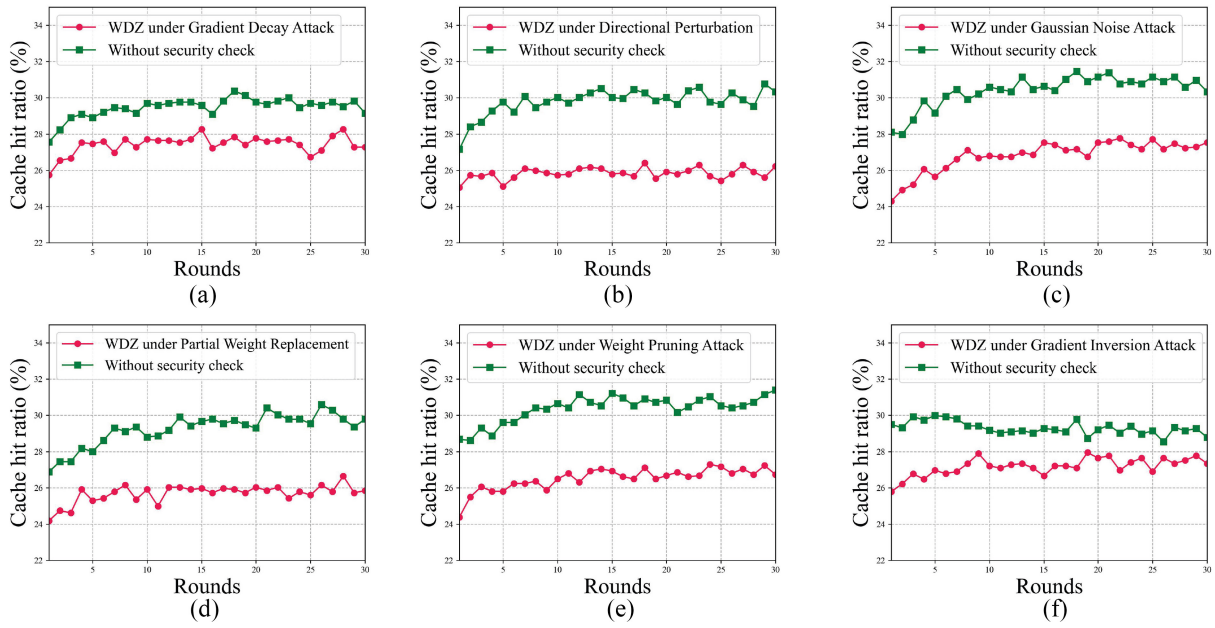


Fig. 10. (a) Gradient Decay Attack. (b) Directional Perturbation. (c) Gaussian Noise Attack. (d) Partial Weight Replacement. (e) Weight Pruning Attack. (f) Gradient Inversion Attack.

so that it is more likely for vehicles to fetch the requested content from the neighboring ENs. Moreover, we can observe that the CHR of both the random scheme and Thompson sampling scheme are lower, as they do not take into account the content popularity prediction that could obtain significant CHR improvement. Additionally, we can note that compared to the n-ε-greedy scheme, the CHR of our proposed SFMA3C scheme is significantly improved. This is attributed to the fact that the proposed SFMA3C integrates the CVAE-GAN model within SAFL to capture effectively underlying patterns and dynamic features between VUs and the local data, thereby predicting accurately interested contents of VUs.

Fig. 7 depicts the CTD of different schemes varies with different caching capacities of vehicles and RSUs. It can be seen that the CTD decreases as the caching capacity increases. This is because that both vehicles and RSUs with larger capacity can cache more contents, and the requested contents are more likely to be fetched from neighboring ENs, thereby shortening the delivery time. Moreover, we can find that the CTD of SFMA3C is lowest among all other baseline schemes. Because the proposed SFMA3C accurately predicts content popularity, enhancing the CHR of ENs. This allows more vehicles to directly retrieve their desired content from the local RSU and neighboring vehicles, thus significantly reducing the transmission time required to obtain content from other ENs.

Fig. 8 shows the CHR and training time for different prediction models, where the caching capacities of vehicle and RSU are set as 20 contents and 50 contents, respectively. We can observe that compared to current prediction models, the CHR of our proposed model is the highest. Especially, compared with the currently popular C-AAE model, the CVAE-GAN model can achieve 7.4% gain of CHR. This is due to the fact that the proposed CVAE-GAN model exhibits superior generative and potential spatial learning capabilities. Specifically, it can generate older samples through iterative training on historical tasks. As a result, the CVAE-GAN model effectively mitigates the issue of model collapse and enhances the prediction accuracy. However, the network structure of the CVAE-GAN model is more complex compared to the AE and its variant, the C-AAE model. Consequently, the CVAE-GAN model requires more computation resources, resulting in slightly longer training times under the same resource conditions. Fortunately, with advancements in computing power, the resources available in RSUs and vehicles are sufficient to fully meet the requirements of the CVAE-GAN model.

We further analyze the effect of the WDZ-based security check on the CHR, as shown in Fig. 9. We assume that during each round of model aggregation, the system may randomly experience one of six different types of malicious attacks: 1) Gradient Decay Attack; 2) Directional Perturbation; 3) Gaussian Noise Attack; 4) Partial Weight Replacement; 5) Weight Pruning Attack; and 6) Gradient Inversion Attack. These attacks occur during the local model upload process. In this case, Fig. 9 demonstrates that the superiority of the WDZ-based method. This indicates that the proposed WDZ-based security check mechanism effectively enhances the model's security and accuracy in the presence of network attacks, making it well-suited for highly dynamic IoV scenarios.

To gain deeper insights into the impact of various attacks on the accuracy of prediction model, Fig. 10 provides a detailed comparison of the WDZ-based security check against six different attack types. Fig. 10(a) to (f) illustrate the model performance under each specific attack scenario. The results clearly demonstrate that the WDZ-based approach significantly enhances the model's robustness and security across all attack scenarios. These findings demonstrate the effectiveness of the WDZ-based security check mechanism, even in the presence of diverse and unpredictable attack conditions.

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

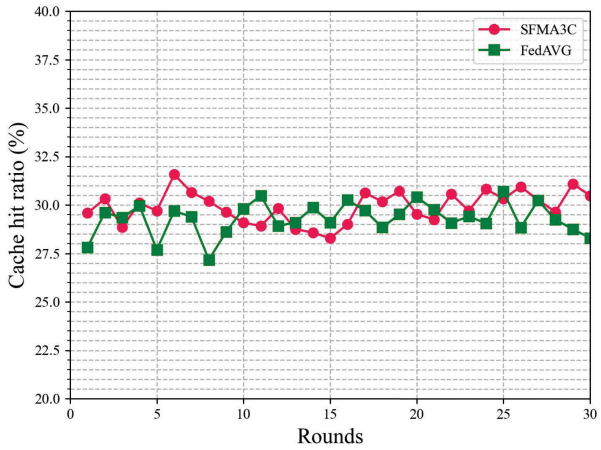


Fig. 11. Comparison of CHR between SFMA3C and FedAVG.

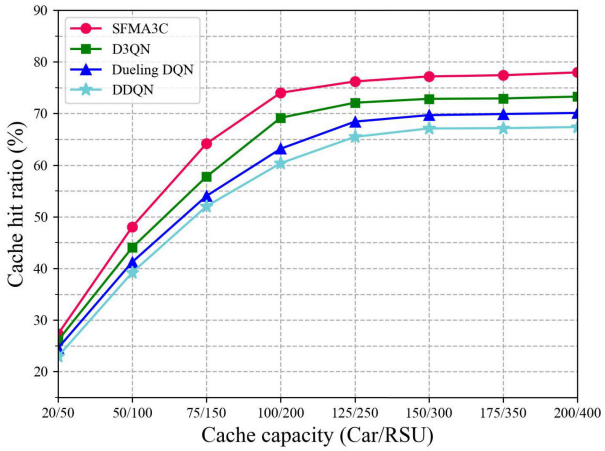


Fig. 13. CHR varies with the cache capacities of vehicles and RSUs.

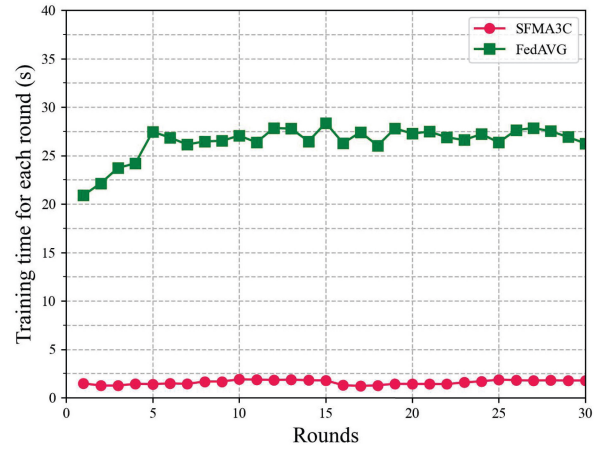


Fig. 12. Comparison of training time between SFMA3C and FedAVG.

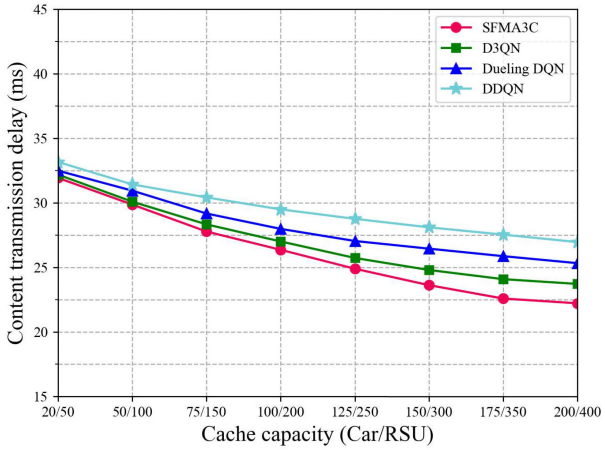


Fig. 14. CTD varies with the cache capacities of vehicles and RSUs.

Fig. 11 compares the CHR performance of the proposed SFMA3C scheme with the FedAVG scheme over 30 rounds, with vehicle and RSU caching capacities set to 20 and 50 contents, respectively. The results show that the CHR of SFMA3C fluctuates between 28% and 32%, slightly outperforming FedAVG, which ranges from 27% to 31%. This highlights the advantage of SFMA3C over FedAVG in terms of CHR. The improvement is attributed to SFMA3C's asynchronous aggregation method. Different from the synchronous approach used in FedAVG, asynchronous aggregation considers vehicle mobility factors, such as velocity and position during the global model aggregation process, thereby enhancing the model's overall performance.

Fig. 12 compares the training time between the SFMA3C and FedAVG schemes across different rounds, with the caching capacities of vehicles and RSU set to 20 and 50 contents, respectively. The results illustrate that the training time per round for SFMA3C fluctuates between 1 s and 2 s, while FedAVG requires significantly longer, with the training time ranging from 21 s and 28 s. These results demonstrate that the SFMA3C scheme significantly reduces training time compared to FedAVG. This improvement stems from the asynchronous nature of SFMA3C, which enables immediate aggregation of local models as they are updated, whereas FedAVG requires waiting for all local models to be aggregated globally. This demonstrates the effectiveness of the SFMA3C method in asynchronous aggregation and further demonstrates its suitability for dynamic IoV environments.

To evaluate the cooperative caching performance of the proposed multiagent A3C approach within the SAFL framework, we compare the proposed SFMA3C with existing schemes in terms of CHR and CTD. Figs. 13 and 14 demonstrate that our proposed SFMA3C scheme outperforms the Dueling DQN [12], DDQN [13], and D3QN [32] schemes in terms of CHR and CTD, respectively. Specifically, it can be observed from Fig. 13 that when the cache capacities of vehicles and RSUs are 75 contents and 150 contents, respectively, the proposed SFMA3C achieves a CHR gain of over 10% compared to D3QN, Dueling DQN and DDQN. Meanwhile, as shown in Fig. 14, the CTD of SFMA3C is reduced by more than 4% compared to the other schemes. This superiority stems from the asynchronous framework of SFMA3C. This framework enables each EN to effectively train its own policy in parallel through multiple agents and

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

make decision independently even under unstable network conditions. This means that the proposed SFMA3C is effective and suitable for the cooperative caching scenarios in highly dynamic IoVs.

VI. Conclusion

In this article, we present a novel mobility-aware cooperative edge caching scheme by integrating SAFL with multiagent A3C, termed SFMA3C, to improve the CHR and reduce the CTD in highly dynamic IoVs. We first presented an innovative SAFL framework that integrates a WDZ-based security check within AFL to ensure model security. Subsequently, To adapt to the dynamic IoV environments, we introduced a proactive caching approach that utilizes CVAE-GAN within SAFL to capture the latent features and enhance the prediction accuracy of popular content. Simulation results have demonstrated the superiority of the proposed CVAE-GAN model. In addition, to maximize the benefits of content prediction within SAFL, we developed a multiagent A3C-based caching decision algorithm to optimize cooperative caching strategies, with the goal of improving CHR and minimizing CTD in highly dynamic IoVs. Simulation results have demonstrated that the proposed SFMA3C scheme outperforms the other schemes in both CHR and CTD. Based on simulation results, the main conclusions can be summarized as follows.

1) The proposed SAFL framework not only enhances model security but also achieves higher CHR in IoVs compared to the conventional AFL framework.
2) Compared to the existing AE and C-AAE models, the proposed CVAE-GAN model obtains higher accuracy in popular content prediction, achieving a CHR improvement of over 7.4% in highly dynamic IoV environments. However, its more complex network structure demands greater computation resources for model training.
3) The proposed SFMA3C outperforms Dueling DQN, DDQN, and D3QN in terms of CHR and CTD. Unlike the synchronous framework adopted by existing schemes, the asynchronous nature of SFMA3C allows each EN to train and make decision in parallel through multiple agents. This design is particularly well-suited for cooperative caching in dynamic IoV environments.

References

[1] P. Arthurs, L. Gillam, P. Krause, N. Wang, K. Halder, and A. Mouzakitis, "A taxonomy and survey of edge cloud computing for intelligent transportation systems and connected vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6206–6221, Jul. 2022.

[2] W. Xu, Z. Yang, D. W. K. Ng, M. Levorato, Y. C. Eldar, and M. Debbah, "Edge learning for B5G networks with distributed signal processing: Semantic communication, edge computing, and wireless sensing," *IEEE J. Sel. Top. Signal Process.*, vol. 17, no. 1, pp. 9–39, Jan. 2023.

[3] L. Su and V. K. Lau, "Data and channel-adaptive sensor scheduling for federated edge learning via over-the-air gradient aggregation," *IEEE Internet Things J.*, vol. 9, no. 3, pp. 1640–1654, Feb. 2022.

[4] K. Jiang, C. Sun, H. Zhou, X. Li, M. Dong, and V. C. M. Leung, "Intelligence-empowered mobile edge computing: Framework, issues, implementation, and outlook," *IEEE Netw.*, vol. 35, no. 5, pp. 74–82, Sep./Oct. 2021.

[5] M. A. Javed and S. Zeadally, "AI-empowered content caching in vehicular edge computing: Opportunities and challenges," *IEEE Netw.*, vol. 35, no. 3, pp. 109–115, Jun. 2021.

[6] J. Zhou, F. Chen, Q. He, X. Xia, R. Wang, and Y. Xiang, "Data caching optimization with fairness in mobile edge computing," *IEEE Trans. Services Comput.*, vol. 16, no. 3, pp. 1750–1762, May/Jun. 2023.

[7] K. Zhang, J. Cao, S. Maharjan, and Y. Zhang, "Digital twin empowered content caching in social-aware vehicular edge networks," *IEEE Trans. Comput. Soc. Syst.*, vol. 9, no. 1, pp. 239–251, Feb. 2022.

[8] Z. Hu, C. Fang, Z. Wang, S.-M. Tseng, and M. Dong, "Many-objective optimization-based content popularity prediction for cache-assisted cloud–edge–end collaborative IoT networks," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 1190–1200, Jan. 2024.

[9] J. Zhu, R. Li, X. Chen, S. Mao, J. Wu, and Z. Zhao, "Semantics-enhanced temporal graph networks for content popularity prediction," *IEEE Trans. Mobile Comput.*, vol. 23, no. 8, pp. 8478–8492, Aug. 2024.

[10] M. Zhao and M. R. Nakhai, "A unified federated deep Q-learning caching scheme for scalable collaborative edge networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 10855–10866, Dec. 2024, doi: 10.1109/TMC.2024.3382824.

[11] Z. Yu, J. Hu, G. Min, Z. Zhao, W. Miao, and M. S. Hossain, "Mobility-aware proactive edge caching for connected vehicles using federated learning," *IEEE Trans. Intell. Trans. Syst.*, vol. 22, no. 8, pp. 5341–5351, Aug. 2021.

[12] Q. Wu, Y. Zhao, Q. Fan, P. Fan, J. Wang, and C. Zhang, "Mobility-aware cooperative caching in vehicular edge computing based on asynchronous federated and deep reinforcement learning," *IEEE J. Sel. Topics Signal Process.*, vol. 17, no. 1, pp. 66–81, Jan. 2023.

[13] K. Jiang, Y. Cao, Y. Song, H. Zhou, S. Wan, and X. Zhang, "Asynchronous federated and reinforcement learning for mobility-aware edge caching in IoVs," *IEEE Internet Things J.*, vol. 11, no. 9, pp. 15334–15347, May 2024.

[14] Z. Qian, Y. Feng, C. Dai, W. Li, and G. Li, "Mobility-aware proactive video caching based on asynchronous federated learning in mobile edge computing systems," *Appl. Soft Comput.*, vol. 162, Sep. 2024, Art. no. 111795.

[15] D. Qiao, S. Guo, D. Liu, S. Long, P. Zhou, and Z. Li, "Adaptive federated deep reinforcement learning for proactive content caching in edge computing," *IEEE Trans. Parall. Distrib. Syst.*, vol. 33, no. 12, pp. 4767–4782, Dec. 2022.

[16] R. Cheng, Y. Sun, Y. Liu, L. Xia, D. Feng, and M. A. Imran, "Blockchain-empowered federated learning approach for an intelligent and reliable D2D caching scheme," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 7879–7890, Jun. 2022.

[17] B. Jedari, G. Premsankar, G. Illahi, M. D. Francesco, A. Mehrabi, and A. Ylä-Jääski, "Video caching, analytics, and delivery at the wireless edge: A survey and future directions," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 431–471, 1st Quart., 2021.

[18] Q. Duan, J. Huang, S. Hu, R. Deng, Z. Lu, and S. Yu, "Combining federated learning and edge computing toward ubiquitous intelligence in 6G network: Challenges, recent advances, and future directions," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2892–2950, 4th Quart., 2023.

[19] Y. Chen, Y. Ning, M. Slawski, and H. Rangwala, "Asynchronous online federated learning for edge devices with non-IID data," in *Proc. IEEE Int. Conf. Big Data*, Dec. 2020, pp. 15–24.

[20] L. Liu, Y. Li, Y. Xu, Q. Zhang, and Z. Yang, "Deep learning-enabled file popularity-aware caching replacement for satellite-integrated content-centric networks," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 5, pp. 4551–4565, Oct. 2022.

[21] X. Bi and L. Zhao, "Collaborative caching strategy for RL-based content downloading algorithm in clustered vehicular networks," *IEEE Internet Things J.*, vol. 10, no. 11, pp. 9585–9596, Jun. 2023.

[22] L. Yao, A. Chen, J. Deng, J. Wang, and G. Wu, "A cooperative caching scheme based on mobility prediction in vehicular content centric networks," *IEEE Trans. Veh. Technol.*, vol. 67, no. 6, pp. 5435–5444, Jun. 2018.

[23] Y. Liu, C. Yang, X. Chen, and F. Wu, "Joint hybrid caching and replacement scheme for UAV-assisted vehicular edge computing networks," *IEEE Trans. Intell. Veh.*, vol. 9, no. 1, pp. 866–878, Jan. 2024.

[24] P. Qin, Y. Fu, J. Zhang, S. Geng, J. Liu, and X. Zhao, "DRL-based resource allocation and trajectory planning for NOMA-enabled multi-UAV collaborative caching 6G network," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 8750–8764, Jun. 2024.

[25] Y.-T. Wang et al., "Markov clustering-based content placement in roadside-unit caching with deadline constraint," *IEEE Trans. Intell. Trans. Syst.*, vol. 25, no. 9, pp. 11881–11892, Sep. 2024, doi: 10.1109/TITS.2024.3368413.

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.

[26] C. Ling et al., "Cooperative service caching in vehicular edge computing networks based on transportation correlation analysis," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 22754–22767, Jun. 2024.
[27] P. He, L. Cao, Y. Cui, R. Wang, and D. Wu, "Multi-agent caching strategy for spatial-temporal popularity in IoV," *IEEE Trans. Veh. Technol.*, vol. 72, no. 10, pp. 13536–13546, Oct. 2023.
[28] X. Li, W. Ouyang, M. Pan, S. Lv, and Q. Ma, "Continuous learning method of radar HRRP based on CVAE-GAN," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5107819.
[29] A. Jaleel, K. B. Theobald, S. C. Steely, and J. Emer, "High performance cache replacement using re-reference interval prediction (RRIP)," in *Proc. Annu. Int. Symp. Comput. Archit.*, Jun. 2010, pp. 60–70.
[30] M. Ahmed, S. Traverso, P. Giaccone, E. Leonardi, and S. Niccolini, "Analyzing the performance of LRU caches under non-stationary traffic patterns," 2013, *arXiv:1301.4909*.
[31] X. Ye, M. Li, P. Si, R. Yang, Z. Wang, and Y. Zhang, "Collaborative and intelligent resource optimization for computing and caching in IoV with blockchain and MEC using A3C approach," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 1449–1463, Feb. 2023.
[32] Z. Zhang, M. St-Hilaire, X. Wei, H. Dong, and A. E. Saddik, "How to cache important contents for multi-modal service in dynamic networks: A DRL-based caching scheme," *IEEE Trans. Multimedia*, vol. 26, pp. 7372–7385, Feb. 2024.
[33] T. Wu, D. Yu, C. Liu, D. Wang, and B. Huang, "Recommendation-enabled edge caching and D2D offloading via incentive-driven deep reinforcement learning," *IEEE Trans. Services Comput.*, vol. 17, no. 4, pp. 1724–1738, Jul./Aug. 2024, doi: 10.1109/TSC.2024.3351219.
[34] T. Wang et al., "Towards intelligent adaptive edge caching using deep reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 23, no. 10, pp. 9289–9303, Oct. 2024, doi: 10.1109/TMC.2024.3361083.
[35] W. Zhang, G. Zhang, and S. Mao, "Deep reinforcement learning based joint caching and resources allocation for cooperative MEC," *IEEE Internet Things J.*, vol. 11, no. 7, pp. 12203–12215, Apr. 2024.
[36] J. Chen, H. Wu, P. Yang, F. Lyu, and X. Shen, "Cooperative edge caching with location-based and popular contents for vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 9, pp. 10291–10305, Sep. 2020.
[37] Y. AlNagar, S. Hosny, and A.-A. El-Sherif, "Towards mobility-aware proactive caching for vehicular Ad Hoc networks," in *Proc. IEEE Wireless Commun. Netw. Conf. Workshop (WCNCW)*, Aug. 2019, pp. 1–6.
[38] C. Liu et al., "Learning-based predictive beamforming for integrated sensing and communication in vehicular networks," *IEEE J. Sel. Areas. Commun.*, vol. 40, no. 8, pp. 2317–2334, Aug. 2022.
[39] F. M. Harper and J. A. Konstan, "The movielens datasets: History and context," *ACM Trans. Interact. Intell. Syst.*, vol. 5, no. 4, pp. 1–19, Feb. 2015.



Xuefang Nie (Member, IEEE) received the Ph.D. degree in information and communication engineering from Harbin Institute of Technology, Harbin, China, in 2018.

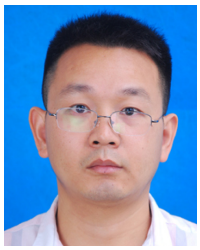
He is currently an Associate Professor with the School of Information Engineering, East China Jiaotong University, Nanchang, China. His current projects are related to cellular V2X communication with artificial intelligence. His research focuses on 5G, C-V2X, wireless communications, stochastic geometry, mobile edge computing, federated learning, and heterogeneous cellular networks including interference management techniques and resource allocation optimization.

Dr. Nie is currently a member of China Computer Federation.



Chen Wang received the B.E. degree in software engineering from Taiyuan university of technology, Taiyuan, China, in 2023. He is currently pursuing the M.S. degrees with East China JiaoTong University, Nanchang, China.

His research interests include deep learning, asynchronous federated learning, and mobile-edge computing.



Tianqing Zhou (Member, IEEE) received the Ph.D. degree in information and communication engineering from Southeast University, Nanjing, China, in 2016.

He joined the School of Information Engineering, East China Jiaotong University, Nanchang, China, in 2016, where he is currently an Associate Professor. His current research interests include mobile edge computing and caching, and artificial intelligence algorithms.



Qiangqiang Zhou (Member, IEEE) received the Ph.D. degree from Tongji University, Shanghai, China, in 2018.

He is currently a Senior Lecturer with the School of Software, Jiangxi Normal University, Nanchang, China. His research interests include pattern recognition, machine learning, and computer vision.



Xusheng Zhu (Graduate Student Member, IEEE) received the B.S. degree in optical communication from Jilin University, Changchun, China, in 2017, and the M.S. degree in electronics and communication engineering from Lanzhou University, Lanzhou, China, in 2022. He is currently pursuing the Ph.D. degree with the Department of Electric Engineering, Shanghai Jiao Tong University, Shanghai, China.

In 2017, he worked as a Network Engineer with Hisense, Qingdao, china. His research interests include index modulation, reconfigurable intelligent surface, and integrated sensing and communication.

Mr. Zhu serves as the TPC Member of the VTC-Fall 2023.



Jiliang Zhang (Senior Member, IEEE) received the B.E., M.S., and Ph.D. degrees from Harbin Institute of Technology, Harbin, China, in 2007, 2009, and 2014, respectively.

He was a Postdoctoral Fellow with Shenzhen Graduate School, Harbin Institute of Technology from 2014 to 2016, an Associate Professor with the School of Information Science and Engineering, Lanzhou University, Lanzhou, China, from 2017 to 2019, and a Researcher with the Department of Electrical Engineering, Chalmers University of Technology, Gothenburg, Sweden from 2017 to 2018. He is currently a Marie Curie Research Fellow with the Department of Electronic and Electrical Engineering, The University of Sheffield, Sheffield, U.K. His current research interests include, but are not limited to wireless channel modelling, modulation system, relay system, wireless ranging system, vehicular communications, ultradense small cell networks, neural dynamic, and smart environment modeling.

Authorized licensed use limited to: NORTHWESTERN POLYTECHNICAL UNIVERSITY. Downloaded on May 20,2026 at 02:41:43 UTC from IEEE Xplore. Restrictions apply.